DOA Algo

Beamforming $K=1$. weight. $w^H a(\theta)$ scan

- CB / Delay-Sum $\hat{\theta}_{CB} = \arg\max\ a^H(\theta)\hat{R}a(\theta)$
- MVDR / Coupon $\hat{\theta}_{MVDR} = \arg\min\ 1/a^H(\theta)\hat{R}^{-1}a(\theta)$

Subspace SVD $\hat{R}$ as $U_s + U_n$

- MUSIC $\hat{\theta}_{MUSIC} = \arg\min\ 1/a^H(\theta)U_nU_n^H a(\theta)$
- ESPRIT

Sparse Model. CS Algo $y = \underset{M\times K}{A(\theta)}\ \underset{K\times 1}{x} = \underset{M\times N}{A} \cdot \underset{N\times 1}{x}$

- Greedy : OMP
- Threshold : HTP. IHT
- Optimization : LASSO $\min \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|x\|_1$

  SR. LAD. Group ...

Parametric, statistical θ estimation

1. Freq : x. θ fixed unknown
   - ML
     - DML / Nonliear LS $\hat{x}_{DML} = A^{\dagger}(\hat{\theta})y$
     - SML $\min\ tr(\hat{R}R^{-1}) + \log|R|$

       $\rightarrow$ SPICE $\min\ tr(\hat{R}R^{-1}) + tr(R\hat{R}^{-1})$
   - BLUE $\hat{x}_{BLUE} = (A^H\Sigma^{-1}A)^{-1}A^H\Sigma^{-1}y$
   - LS $\hat{x}_{LS} = A^{\dagger}y$
   - EM CRLB analysis for Freq
2. Bayes : x is RV. $x \sim N(0, P)$
   - MMSE / SBL $\hat{x}_{MMSE} = PA^HR^{-1}y$
   - MAP